ENGINE CONTROL UNIT · KIRLOSKAR 3GK550ETA 4SR1

Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

24 blocks all checks **passing** generated from **sim/blocks/** ← Field handbook

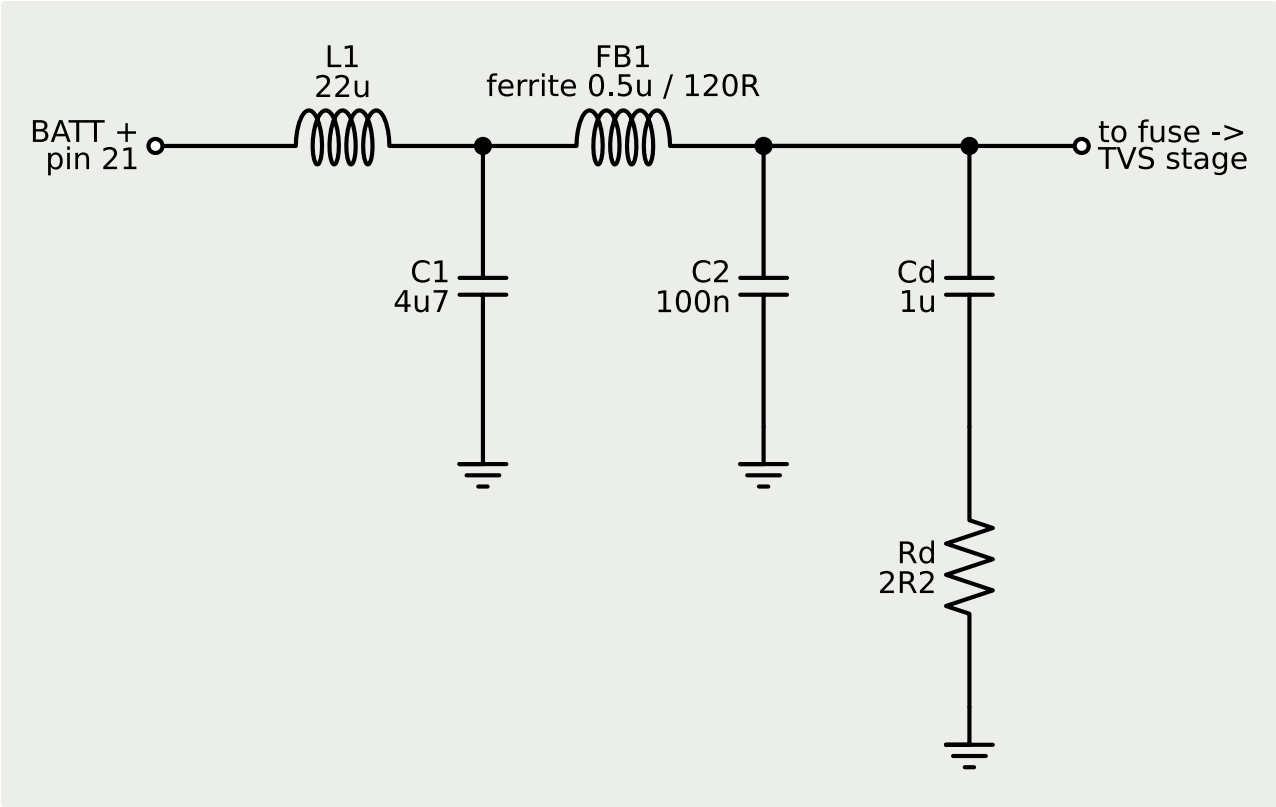
Sensor reference →

01 Power chain
02 Sensor front-ends
03 Speed and position
04 Injection
05 Actuators
06 Communications

01 Power chain

From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.

EMI input filter `pin 21` `10/10 checks`



Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

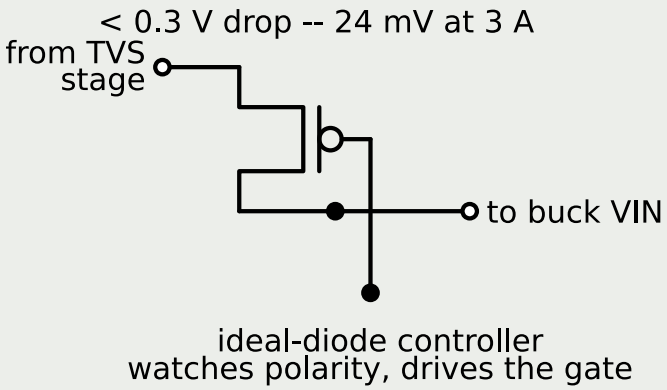
Why it looks like that. One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either — without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

> **What the model does not cover.** Differential mode only. Common-mode noise — both wires swinging together against chassis, which is most real automotive failures — needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

PASS insertion loss at 150 kHz (band start)		-41.071 dB
PASS insertion loss at 400 kHz (switching f)		-51.546 dB
PASS insertion loss at 1 MHz		-74.812 dB
PASS insertion loss at 30 MHz		-141.697 dB
PASS insertion loss at 100 MHz (FM band)		-128.977 dB
PASS worst insertion loss anywhere above 150 kHz		-41.171 dB
PASS ... worst point sits at		151.5 kHz
PASS is 40 dB enough for Class 5?	unknown -- needs measured unfiltered emissions on hardware	
PASS ... IL@150kHz, C1 at X7R's own -15% temp derate (C1=3.995uF)		-39.736 dB
PASS ... IL@400kHz, same corner		-50.070 dB

Netlist: sim/blocks/emi_filter.cir

Reverse-battery protection – 8/8 checks



A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

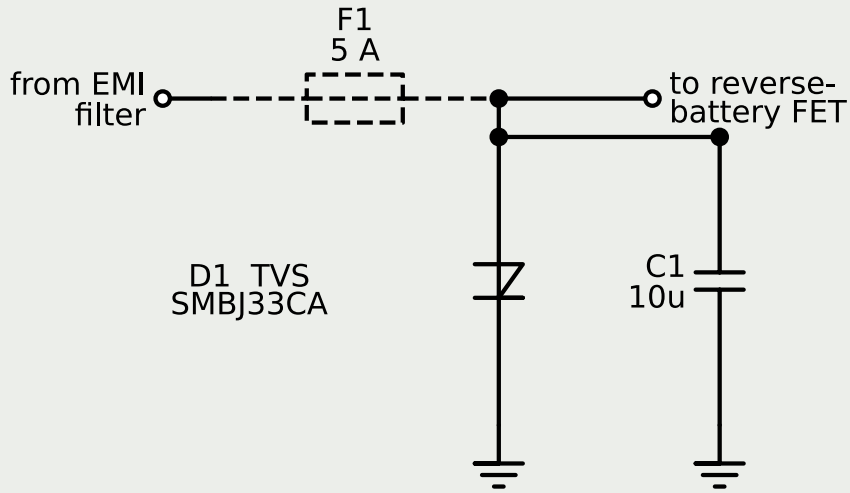
Why it looks like that. Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

> **What the model does not cover.** Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

PASS	Schottky drop at 3 A	0.465 V
PASS	ideal-diode FET drop at 3 A	0.024 V
PASS	FET drop stays under 0.3 V spec to 5 A	0.040 V
PASS	... Schottky does NOT meet it	0.52 V at 5 A -- fails the 0.3 V spec
PASS	power burnt in the Schottky at 3 A	1.395 W
PASS	... and in the FET instead	0.072 W
PASS	Schottky blocks -14 V (leak into 1k)	1.000 mV
PASS	ideal-diode FET blocks -14 V	0.014 mV

Netlist: sim/blocks/reverse_battery.cir

Transient clamp `pin 21` `6/6 checks`



F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

Why it looks like that. Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

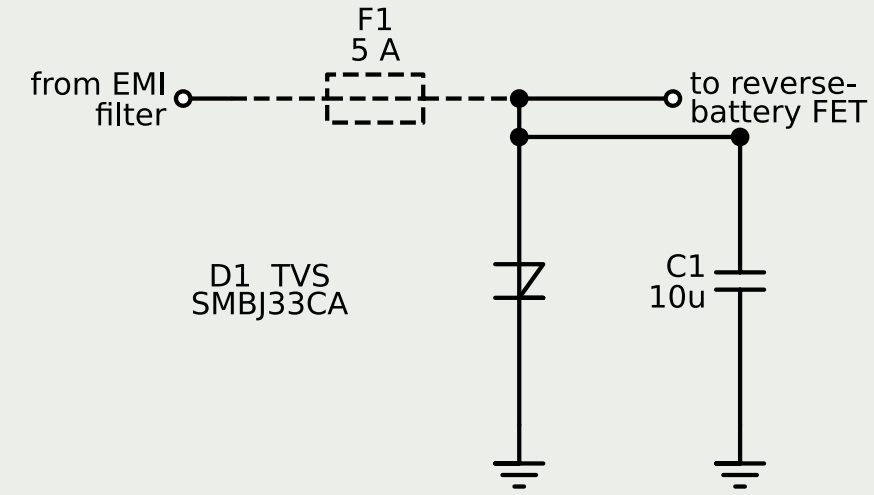
> **What the model does not cover.** It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load_dump, which did check this for its own pulse.

PASS pulse actually applied (source peak)		125.500
PASS clamped voltage at buck input		73.281 V
PASS ... margin inside LM5164's 100 V rating		1.365 x
PASS ... 60 V-class parts are NO LONGER viable downstream		73.281 V
PASS ... so the reverse-battery FET needs 100 V class, not 60	73.3 V clamp -- reverse_battery.cir's 60 V figure is superseded	
PASS recovers to nominal after pulse		12.981 V

Netlist: sim/blocks/transient_clamp.cir

Load dump pin 21

11/11 checks



pulse 5b: Us*=40 V, 0.5R, 400 ms
D1 absorbs ~46 J at ~116 W -- FAILS an SMBJ-class part's rating
F1 dashed (drawn, not in this netlist) does not rescue it either:
fault current is 3.04 A, 0.6x F1's rating, and won't clear in 400 ms

The same TVS and capacitor as transient_clamp, against ISO 7637-2 pulse 5b: a suppressed load dump, 40 V through 0.5 ohms for 400 ms — four orders of magnitude longer than pulse 2a.

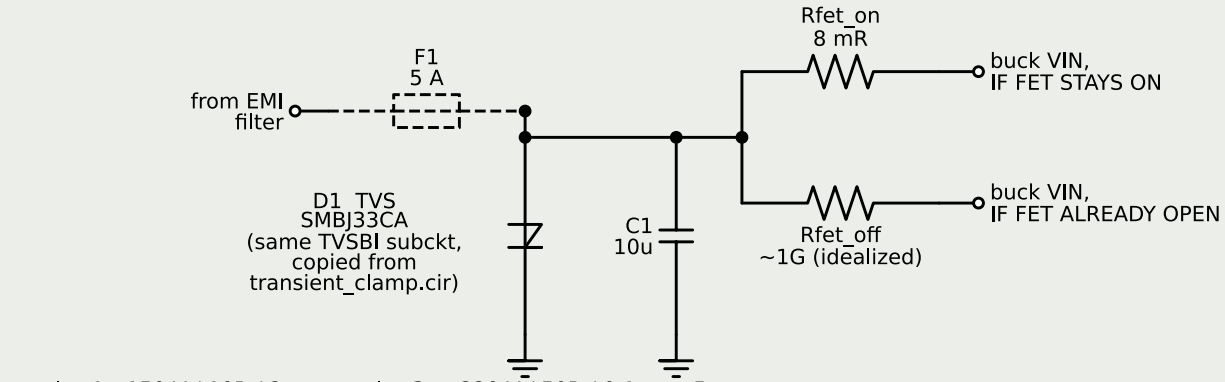
Why it looks like that. Clamps to 38 V, comfortably inside every voltage rating in the chain — the same story as pulse 2a. But 400 ms is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly 46 J at ~116 W average, which is on the order of 100x an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at Us* = 35 V, the figure usually quoted for a 12 V suppressed dump, and the TVS never reaches its 36.7 V breakdown — 0.000 A, 0 J, no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated 100 V, rides 40 V without complaint. Energy absorption only becomes unavoidable under pulse 5a.

> **What the model does not cover.** Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

PASS pulse actually applied (source peak)	40.000 V
PASS clamped voltage through the 400 ms plateau	39.604 V
PASS ... inside LM5164's 100 V rating	2.525 x
PASS ... 60 V-class parts still viable	39.604 V
PASS TVS current through the plateau -- now ZERO, the TVS stays off	0.000 A
PASS average power dissipated in the TVS	0.000 W
PASS total energy absorbed by the TVS over the pulse	0.000 J
PASS energy stays within a single-pulse (~0.5 J) rating	0.000 J
PASS average power stays within continuous (~3 W) rating	0.000 W
PASS ... fraction of the continuous rating now used	0.000 x
PASS recovers to nominal after pulse	13.366 V

Netlist: `sim/blocks/load_dump.cir`

Negative transients `pin 21` `17/20 checks`



The same TVS and bulk capacitor as transient_clamp/load_dump, against ISO 7637-2's two worst NEGATIVE pulses: pulse 1 (−150 V through 10 Ω for 2 ms, a supply disconnection from an inductive load) and pulse 3a (−220 V through 50 Ω for 0.1 ms, simulated as a 5-pulse burst at the standard's own 90 ms minimum repeat). Both Level IV, both sourced from research memo 06's own severity table. This became answerable only once the TVS stopped being modelled as a single diode, which conducts forward — and silently wrong — in reverse polarity.

Why it looks like that. The buck's own datasheet (LM5164, Absolute Maximum Ratings, fetched directly) rates VIN at −0.3 V to 100 V — the negative limit is not a mirror of the positive one. With the reverse-battery FET assumed to still be conducting — the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide — the buck's VIN pin sees −40.7 V on pulse 1 and −27.8 V on pulse 3a: 90–135× past that −0.3 V rating, not a thin margin. The reverse-battery stage does nothing to prevent this on the timescale these pulses occupy; it would help enormously if the controller reacts fast enough, and nothing here — or in reverse_battery.cir — says whether it does. Pulse 1's TVS energy (0.707 J single event) exceeds load_dump's own ~0.5 J class bound by 41%, a second genuine failure, not a manufactured one — TOLERANCE shows it flips to a PASS at Level III instead of IV, the same one-inequality shape load_dump's own Us* finding has. Pulse 3a's bulk capacitor (C1) does something transient_clamp never needed to consider: its own RC (500 µs) is five times the pulse's 90 µs flat top, so C1 arrests the excursion before the TVS ever reaches its 36.7 V breakdown at nominal value — the TVS

carries zero current, and nothing accumulates across the 5-pulse burst. TOLERANCE shows this is fragile: the same MLCC DC-bias derating emi_filter.cir already documents (up to 60% loss) is enough to flip the TVS into conducting at −38.4 V instead of −27.8 V — still a small energy (4.2 mJ/pulse, far under any rating), but a different regime, and the buck sees more negative voltage either way.

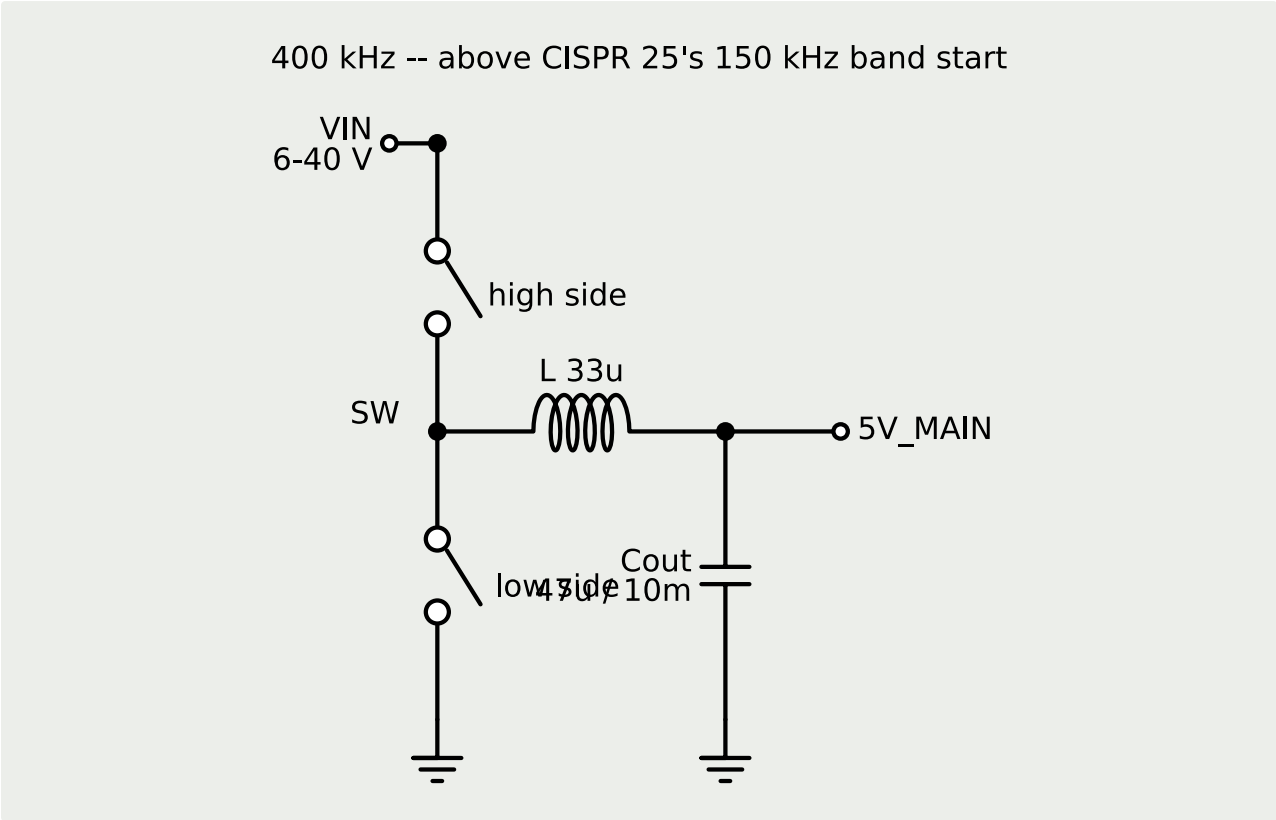
> **What the model does not cover.** The FET's actual turn-off delay is not modelled anywhere in this repo — reverse_battery.cir's own RESULT NOTE says its switch model 'will always look perfect in reverse' because it has none, and no controller part is chosen to pull one from a datasheet. Both bookends (FET fully on, FET already open) are simulated directly rather than guessing at a number in between. The burst is 5 repeats, not the standard's ~40000 (1 hr at 90 ms minimum repeat); nothing here rules out a slow thermal drift that would only show up over the full count, and TVS junction temperature is not modelled at all, the same gap load_dump's own RESULT NOTE states for its much larger energy question. Pulse 1's real ISO waveform is a decaying transient, not the flat top used here — pessimistic in the right direction, same simplification load_dump uses for pulse 5b.

PASS pulse 1 actually applied (source peak)	
PASS pulse 3a actually applied (source peak)	
PASS battery node, pulse 1 -- now set by the clamp diode, not the TVS	
PASS buck VIN pin, pulse 1, WITH the negative clamp fitted	
FAIL ... vs LM5164's OWN -0.3 V negative abs max (sourced, not INFERRED)	
PASS ... residual overshoot past that rating	
PASS battery node, pulse 3a -- now set by the clamp diode, not the TVS	
PASS buck VIN pin, pulse 3a, WITH the negative clamp fitted	
FAIL ... vs LM5164's OWN -0.3 V negative abs max	
PASS ... residual overshoot past that rating	
PASS ... comparison only: IF the FET had already opened (idealized, reverse_battery.cir's own optimistic model, not verified against any real turn-off delay)	
PASS ... same comparison, pulse 3a	
PASS ... so the reverse-battery stage helps only if it reacts in time, and nothing sources that time	FET-on branch tracks the bare TVS clamp to within

PASS bat1 recovers to its own Rsrc1/Rload divider baseline (11.25 V, not 13.5 V)		
PASS bat3a recovers to its own divider baseline (6.75 V) between every burst pulse		
PASS burst: peak-to-peak spread across all 5 repeats (accumulation would show up here)		
FAIL TVS energy, pulse 1 (single event, Level IV)		
PASS ... vs load_dump's own ~0.5 J single-pulse class bound		
PASS TVS energy, pulse 3a (nominal C1 -- TVS stays below its own breakdown, does not conduct)		
PASS ... so nothing here can accumulate at nominal C1 -- see TOLERANCE for the corner where it would	TVS current is numerically zero throughout the burst at C1	

Netlist: sim/blocks/negative_pulses.cir

Buck pre-regulator – 8/8 checks



6–40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

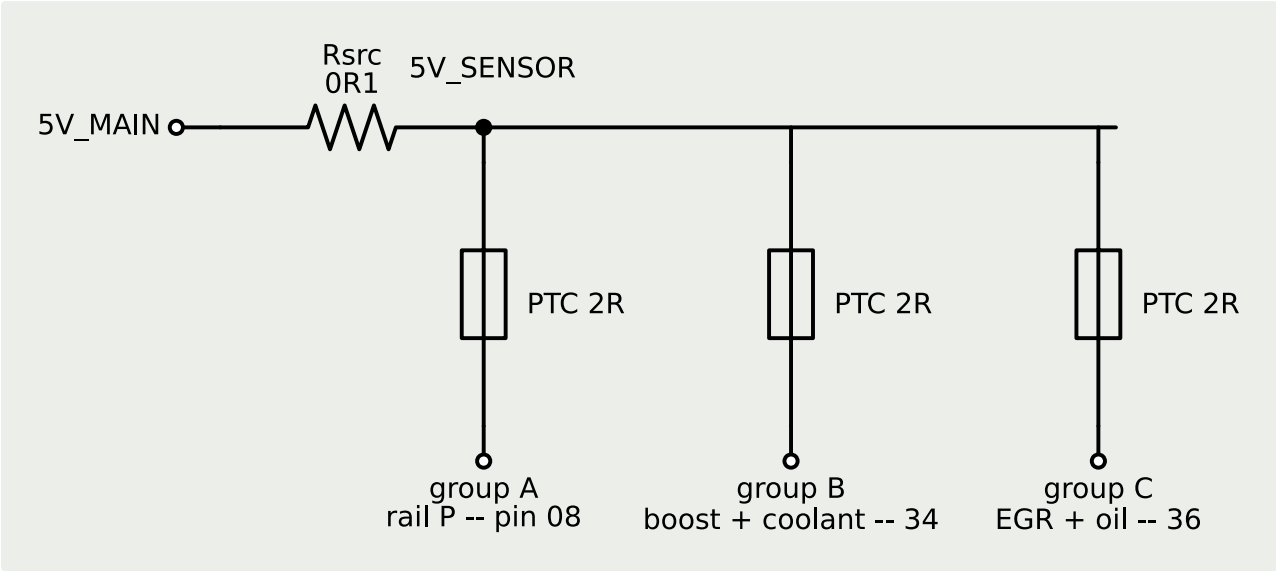
Why it looks like that. 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low — ripple current grows with input voltage even though duty cycle falls.

> **What the model does not cover.** Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

PASS output at 13.5 V in (open loop, D=0.370)	4.926 V
PASS output at 40 V in (open loop, D=0.125)	4.945 V
PASS inductor ripple at 13.5 V	0.256 A
PASS inductor ripple at 40 V	0.348 A
PASS ... worst case is HIGH line, not low	1.362 x
PASS worst-case ripple as fraction of 1 A load	34.840 %
PASS output ripple at 13.5 V	13.845 mV
PASS output ripple at 40 V	21.423 mV

Netlist: sim/blocks/buck_preregulator.cir

5V_SENSOR distribution – 6/7 checks



One resettable fuse per sensor group, and the same fault simulated without them.

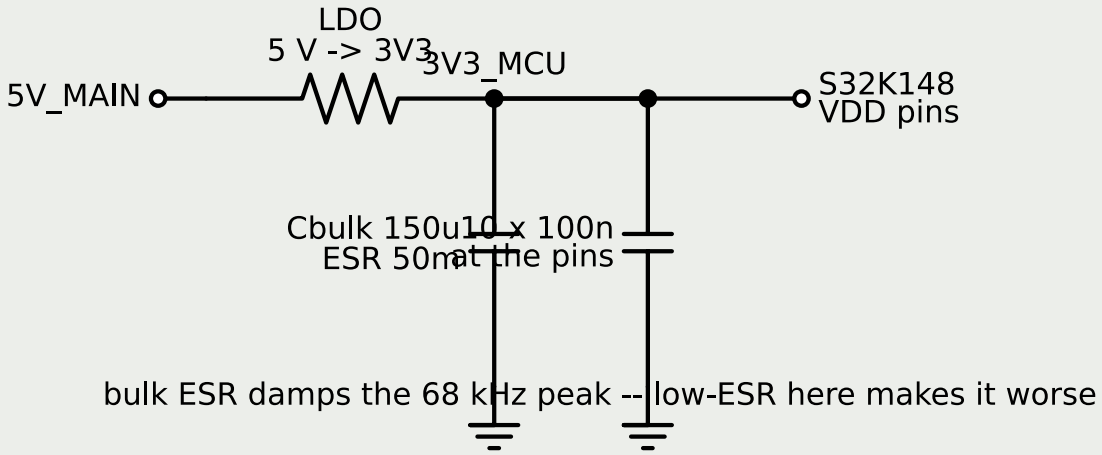
Why it looks like that. A 5 V sensor supply shorted to engine ground is a routine fault, not an exotic one. The question is what it does to the OTHER sensors: with per-group protection the healthy groups keep 4.66 V, without it the whole rail collapses to 0.5 V and the ECU loses every analog channel at once — which means it cannot tell a wiring fault from an engine failing.

> **What the model does not cover.** The polyfuse is modelled cold, which is the worst case for the healthy groups: the instant after the short, before the device has heated and tripped. Once tripped the healthy groups recover further. Firmware still has to notice — a tripped fuse and a dead sensor look identical from the ADC's side.

PASS healthy group with PTCs, group A shorted		4.661 V
PASS ... rail itself holds up		4.754 V
PASS healthy group WITHOUT PTCs, same fault		0.495 V
PASS ... so one harness short blinds the whole ECU	0.50 V -- every analog channel lost at once	
PASS protection improves the healthy group by		9.409 x
PASS fault current drawn from the 5 V rail		2.458 A
FAIL ... stacked -40C-cold-ambient + -30%-tolerance PTC corner		4.148 A

Netlist: sim/blocks/sensor_rail.cir

3V3_MCU decoupling – 8/9 checks



The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

Why it looks like that. The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz — the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position. It was pulled a second time on 18 September, from 47 uF to 150 uF: at 47 uF the block cleared its target only at the nominal value of the LDO's output inductance, a number no chosen part pins down, and a 26% margin that evaporates when an unmeasured parameter is restated is not a margin. At 150 uF that resonance has dropped and damped enough that a different peak is now the tallest — the bulk/ceramic anti-resonance at 2.6 MHz, floored near 60 mOhm — and that one does not move with the LDO at all. The verdict has stopped depending on the unpinned number. More bulk beyond 150 uF buys nothing, because it does not touch the peak that now dominates.

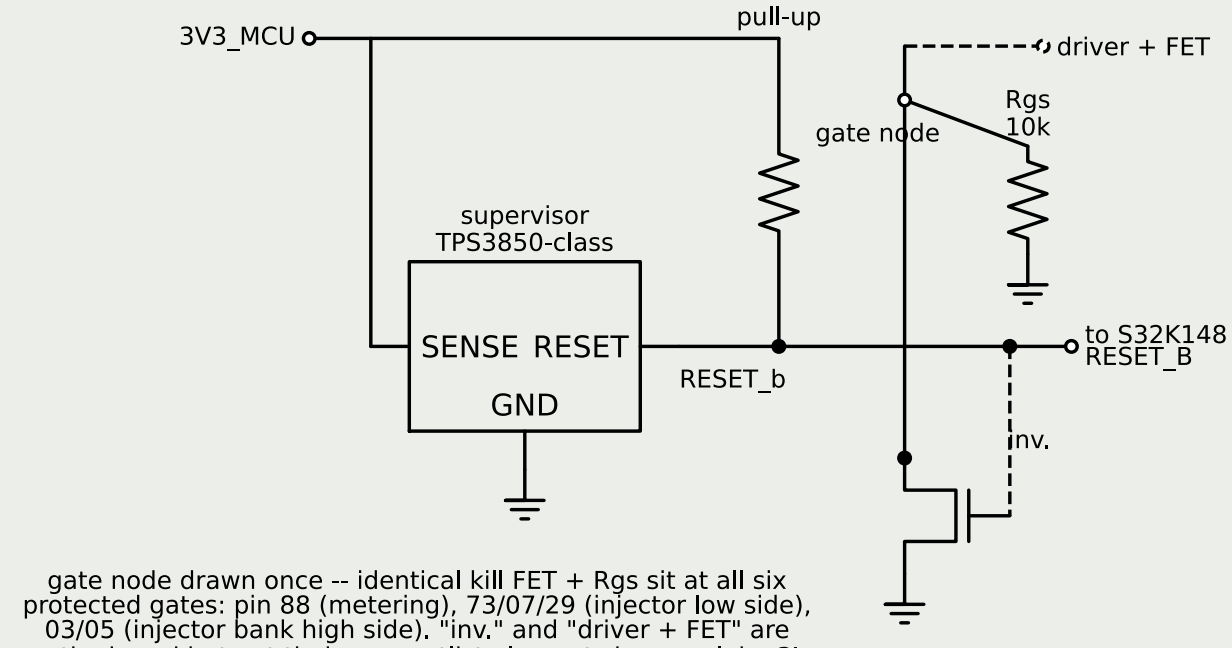
> **What the model does not cover.** Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz — which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

PASS peak impedance, 1 kHz to 100 MHz		60.355 mOhm
PASS ... and it has moved to the MHz anti-resonance		2.630 MHz
PASS ... i.e. bulk against ceramic, no longer regulator L against bulk	2.63 MHz -- the LDO resonance is no longer the limiter	
PASS ten 100 nF parts at 100 MHz		46.923 mOhm
PASS one 1 uF part at 100 MHz		357.421 mOhm
PASS ... same capacitance, this much worse		7.617 x
PASS ten-part net at the 3.5 MHz anti-resonance		55.862 mOhm
PASS ... one-part net is LOWER there (its ESR damps it)	34 mOhm vs 56 -- more ESR, better damping	
FAIL ... peak at a plausible cold bulk-ESR corner (Rb=200 mOhm, 4x)		196.700 mOhm

Netlist: sim/blocks/mcu_pdn.cir

Supervisor / fail-safe kill path –

33/33 checks



gate node drawn once -- identical kill FET + Rgs sit at all six protected gates: pin 88 (metering), 73/07/29 (injector low side), 03/05 (injector bank high side). "inv." and "driver + FET" are on the board but not their own netlist elements here -- claim 3's FAIL (see RESULT NOTE) is about Rgs, drawn and modelled as 10k.

A TPS3850-class windowed supervisor watching 3V3_MCU — the same rail mcu_pdn.cir characterizes — whose RESET drives, through one inverting stage, a kill FET at every protected driver gate (pin 88 metering, 73/07/29 injector low side, 03/05 injector bank high side), in parallel with a standing 10 kΩ gate-source pulldown.

Why it looks like that. Research memo 11 found this board has no external watchdog, no brownout supervisor, and nothing defining output state during MCU reset — and that spec §3's other lines of defence both collapsed on 18 Sep 2026 (the fitted DSE4522 takes engine speed from this ECU over CAN, and its fuel relay signals rather than removes power), making this the first line, not defence-in-depth. Five falsifiable claims came out of that memo; two mattered most. Claim 4, the central one, PASSES with a wide margin: across six brownout ramps spanning 293 µs to 1 s, the kill path always finishes (gate below Vgs(th), fought by an adversarial driver the whole time) before 3V3_MCU crosses 2.97 V — the S32K148's own PLL-guarantee floor — with margin that GROWS at slower ramps, so the memo's flagged-open 'slow sag' case does not threaten it. Claim 3 FAILS, and not narrowly: swept across the memo's own illustrative Crss range under this design's own 2.8 V/µs boost-rail turn-off edge, the 10 kΩ pulldown alone lets the coupled gate voltage exceed a representative Vgs(th) starting at Crss = 50 pF — the LOW end of that range, not an extended one — and the memo's own hand arithmetic claiming '~5.6 mV' at 200 pF is off by three orders of

magnitude (560 µA through 10 kΩ is 5.6 V). A 300 Ω–1 kΩ pulldown clears it with real margin; the fix is a component VALUE, not a new part.

> **What the model does not cover.** The pin-mux gap is real and not papered over: which physical PTxx pins 88/73/07/29/03/05 land on is still a Phase-2 item, so their individual reset pull-state defaults cannot be stated — this model treats every protected gate generically, with no GPIO drive assumed at all except as a deliberately adversarial stimulus. The supervisor IC's own comparator propagation delay (assert side) is not sourced anywhere and is modelled as zero; claim 4's tightest margin (18 µs, at the fastest ramp) is the one that delay would eat into first. The inverting stage and kill FET's own gate-drive dynamics are collapsed into one switch with an INFERRED Ron. Ciss=2 nF and Vgs(th)=1.0 V are class figures, not a chosen part's datasheet numbers — no injector/metering FET has been selected yet.

PASS claim 1: gate below Vgsth, kill engaged (bound: 10-40 us)
PASS ... actual kill-path discharge time
PASS claim 1b: gate stays below Vgsth despite an adversarial driver actively holding it high (GPIO-independence)
PASS claim 2: final V(gate) with the 470 ohm pulldown fitted
PASS ... vs the same driver with no pulldown
PASS ... driver output impedance needed to still reach 9.5 V
PASS ... the 100 ohm driver this block assumed is too weak by
PASS claim 3 (evidence): peak V(gate), Crss=20 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=50 pF, Rpd=10k

PASS claim 3 (evidence): peak V(gate), Crss=100 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=200 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=500 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=1000 pF, Rpd=10k
PASS ... memo 11's own hand arithmetic at Crss=200 pF claimed ~5.6 mV -- off by 1000x (560 uA * 10 kOhm = 5.6 V, not 5.6 mV); thi
PASS ... so the 10k pulldown fails even at the memo's OWN low end
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=293 us
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=585 us
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1.17 ms

PASS claim 4: gate safe before 2.97V guarantee floor, ramp=10 ms
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=100 ms
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1000 ms
PASS ... tightest margin is at 18 us at 293 us -- the FASTEST ramp tried
PASS claim 5: kill-path latency vs one 26.67 ms cylinder interval
PASS ... caveat: excludes the watchdog's not own fault-DETECTION time
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=220 ohm
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=470 ohm
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=1000 ohm

```
PASS claim 6
(sweep): peak
V(gate) at
Crss=500 pF,
Rpd=2200 ohm
```

```
PASS claim 6
(sweep): peak
V(gate) at
Crss=500 pF,
Rpd=4700 ohm
```

```
PASS ... same
sweep read at
Vgsth=2.5 V,
the
permissive
end
```

```
PASS ...
largest
pulldown that
holds the
gate below
Vgsth
```

```
PASS ... cost
of 470 ohm:
drive current
per gate held
at 10 V
```

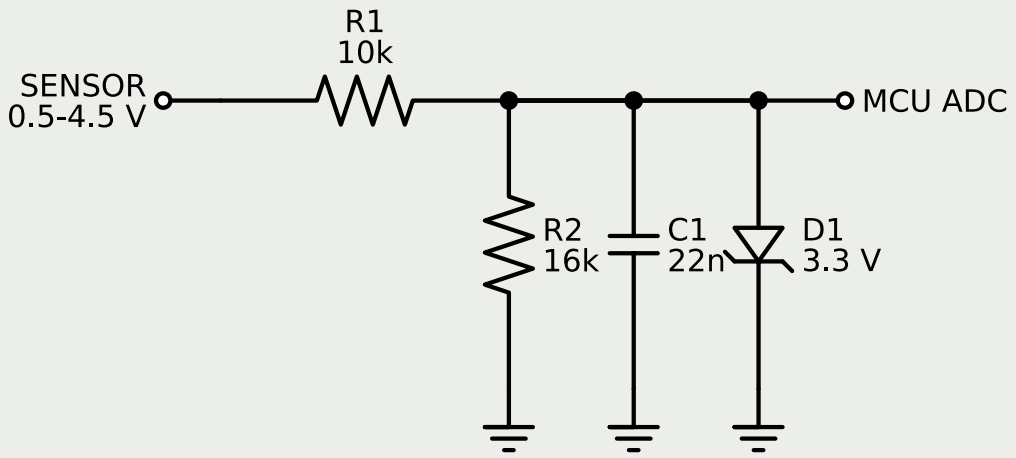
```
PASS ... and
across all
six driver
gates
```

Netlist: sim/blocks/supervisor.cir

02 Sensor front-ends

Everything that turns something on the engine into a number in the ADC.

Ratiometric sensor input pins 41, 35, 80, 37 14/14 checks



The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

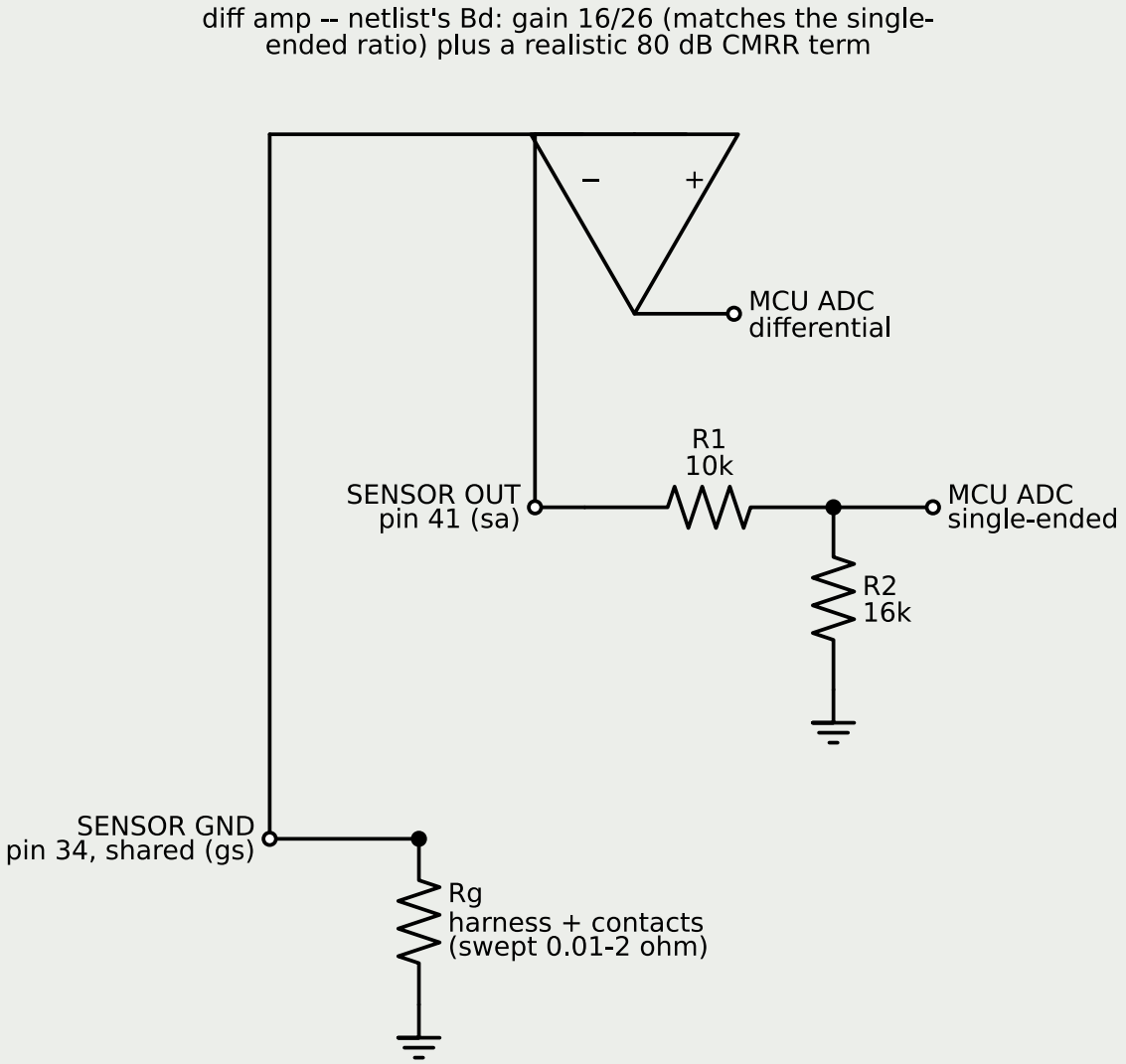
Why it looks like that. Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

> **What the model does not cover.** Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

PASS sensor zero (0.5 V) maps to		0.308 V
PASS sensor full scale (4.5 V) maps to		2.769 V
PASS ... clears the 3.3 V clamp		0.531 V
PASS span used of ADC range		74.592 %
PASS low-pass corner		1,174.898 Hz
PASS loss at sensor's own 100 Hz		-0.031 dB
PASS attenuation at 100 kHz switching		-38.596 dB
PASS 40 V harness short clamps pin to		3.308
PASS ... fault current limited to		3.669 mA
PASS raw reading moves with a +/-5% rail drift		10.000 %
PASS ... so an assumed 5.000 V rail is a 10% error band	10.0% of reading, and nothing in the raw value reveals it	
PASS corrected reading is flat		0.000 %
PASS correction recovers the true fraction		0.500
PASS ... divider ratio cancels, so its tolerance drops out too	measurand = Vsig/Vrail -- k appears in both and divides away	

Netlist: sim/blocks/sensor_ratiometric.cir

Differential sensor input `pin 34 group` `7/7 checks`



Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

Why it looks like that. Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against (+/-2 V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider (R1/R2, the single-ended path) plus a generic amplifier for the differential path,

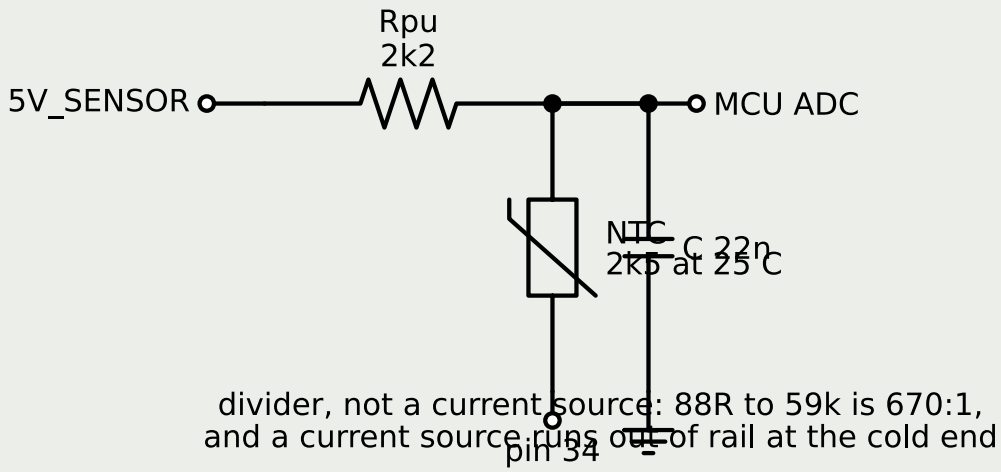
because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

> **What the model does not cover.** The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

PASS clean harness (50 mOhm), single-ended error	0.612 mV
PASS 1 ohm return, ground offset present	19.903 mV
PASS ... single-ended reading error	12.248 mV
PASS ... differential reading error	0.127 mV
PASS differential is better by	96.448 x
PASS single-ended error at 2 ohm (bad contact)	0.995 % span
PASS ... single-ended needs contact resistance below	~0.1 ohm forever, to hold 0.1% -- differential needs nothing

Netlist: `sim/blocks/sensor_differential.cir`

NTC temperature input `pin 79` `11/11 checks`



The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full –40 to 150 C range.

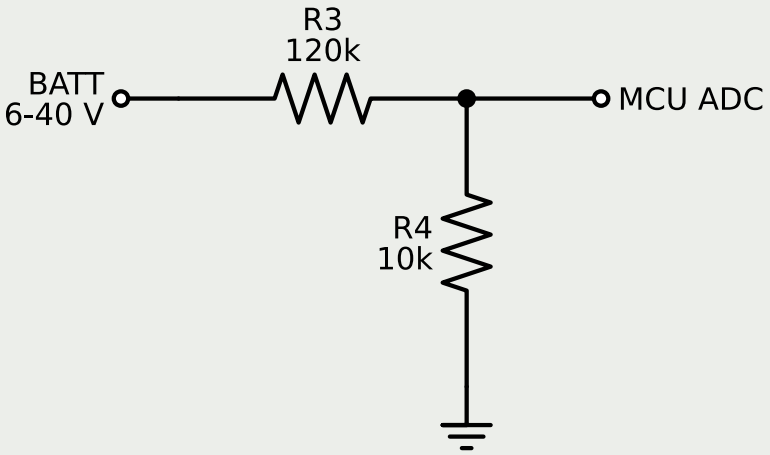
Why it looks like that. Memo 09 found DSE drives its temperature senders with a fixed 15 mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer: 88 ohms to 59 kilohms is a 670:1 span against the 50:1 a genset sender covers, and a current source sized to give useful signal at 150 C demands 5.9 V from a 5 V rail at –40 C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

> **What the model does not cover.** Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

PASS divider at 150 C (88 ohm)		0.192 V
PASS divider at 25 C (2.5k)		2.664 V
PASS divider at -40 C (59k)		4.820 V
PASS ... never leaves the 5 V rail		4.820 V
PASS 100 uA source needs this at -40 C		5.899 V
PASS ... but the rail is 5 V	5.9 V demanded -- out of compliance, reading clips	
PASS 100 uA source at 150 C gives only		8.800 mV
PASS current that WOULD stay in compliance		84.746 uA
PASS ... leaving this at 150 C		7.458 mV
PASS divider self-heating power at 25 C		2.829 mW
PASS ... error at ~2 mW/C dissipation constant		1.414 degC

Netlist: sim/blocks/ntc_frontend.cir

Battery voltage sense `pins 04, 06` `9/9 checks`



R3/R4: the adopted divider (spec sec.4, pins 04/06). battery_sense.cir also carries a rejected R1/R2 75k/10k divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

Why it looks like that. The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

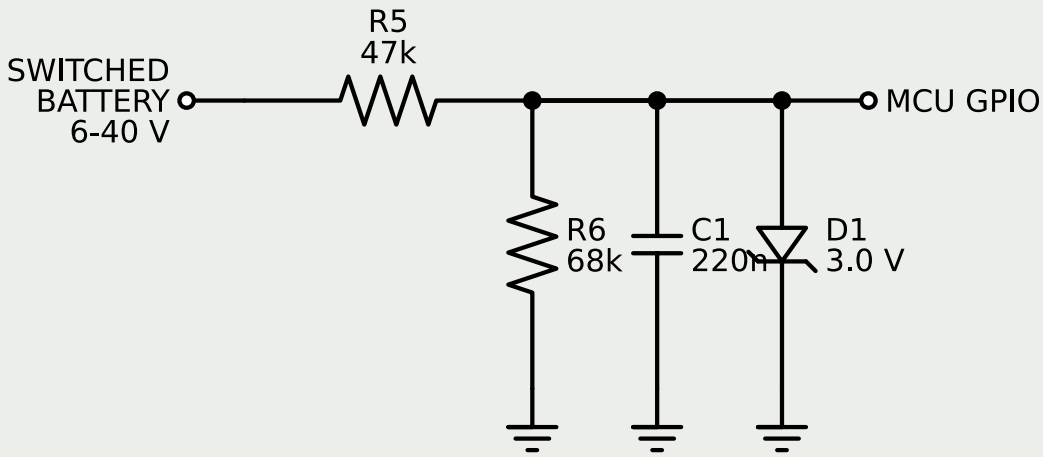
> **What the model does not cover.** Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

PASS	inherited 75k/10k peak at 40 V	4.694 V
PASS	... and so exceeds a 3.3 V ADC	4.69 V > 3.3 V -- would damage input
PASS	corrected 120k/10k peak at 40 V	3.069 V
PASS	... stays under the 3.3 V rail	3.069
PASS	headroom below rail	0.231 V
PASS	reading at 6 V cranking dip	0.462 V
PASS	... resolves to usable ADC counts	572.867 LSB
PASS	divider is linear (max residual)	0.000 V
PASS	slope matches 10/130	0.077

Netlist: sim/blocks/battery_sense.cir

Discrete switch input pins 20, 24, 71

7/7 checks



The switched-battery inputs, active high.

Why it looks like that. This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

> **What the model does not cover.** The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

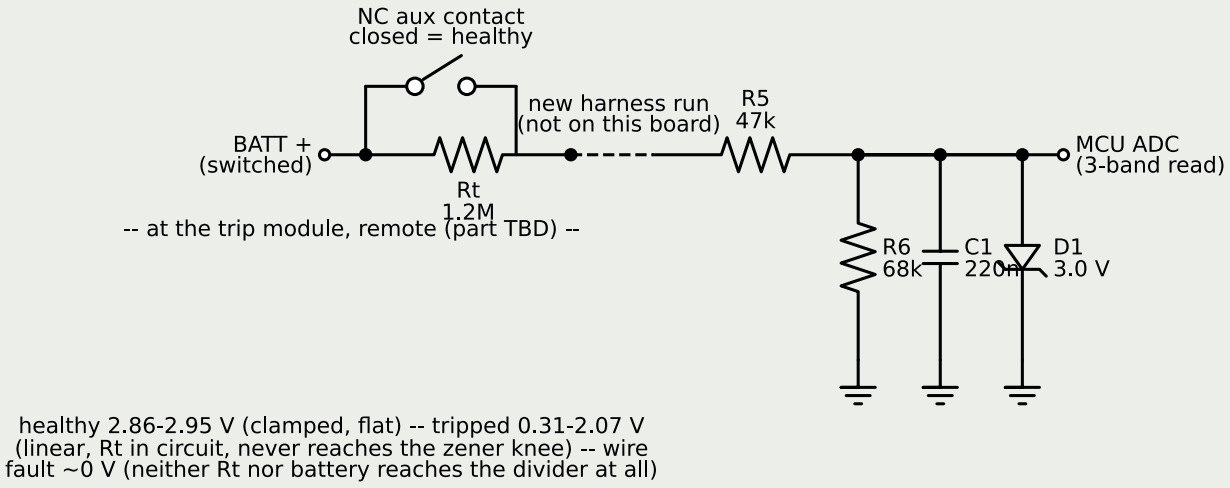
PASS never exceeds 3.6 V abs max over 6-40 V	2.954
PASS logic high at 6 V cranking dip	2.863
PASS logic high at 40 V load dump	2.954
PASS level is flat across 6-40 V (spread)	0.092 V
PASS settled high after bounce ends	2.910
PASS threshold crossings during bounce	0.000
PASS debounce time constant	6.114 ms

Netlist: sim/blocks/discrete_input.cir

Trip-module sense

new pin, no OEM harness

14/14 checks



healthy 2.86-2.95 V (clamped, flat) -- tripped 0.31-2.07 V (linear, Rt in circuit, never reaches the zener knee) -- wire fault ~0 V (neither Rt nor battery reaches the divider at all)

A three-state supervised loop reading whether the mandatory standalone overspeed trip module's contacts are open or closed — not a copy of the input above, on purpose.

Why it looks like that. Spec §3: U12 closed badly on 18 Sep 2026 — the fitted DSE4522's fuel relay only signals, it does not remove power, so a standalone trip module in series with the fuel metering unit's own battery feed (fuse 8F1) is mandatory again, and this ECU must sense its state rather than read a trip as a fuelling anomaly. discrete_input.cir's own accepted limitation — an open wire reads identically to 'not asserted', accepted there because changing the sense would break OEM harness compatibility — does not carry over: this pin has no OEM harness to stay compatible with, and an undetectable broken wire here would read 'not tripped' at the exact moment a trip occurs, which is the one failure spec §3 exists to catch. So this block adds an end-of-line resistor (Rt, 1.2 MΩ) in parallel with the module's own auxiliary contact, physically at the module end of the new harness run, giving three distinguishable ADC bands on the same R5/R6/zener front end discrete_input.cir already validates: healthy (contact closed, Rt shorted) clamps to 2.86–2.95 V exactly as that file's input does; tripped (contact open, Rt in circuit) is a plain, unclamped linear divider that stays 0.31–2.07 V across 6–40 V, never reaching the zener's knee; wire fault (a break anywhere in the run removes both the contact and Rt the same way) reads ~0 V. A contact-bounce transient, reusing discrete_input.cir's own nasty bounce profile inverted, shows the reading never leaves the healthy band during the bounce itself — C1's

time constant on this branch is long enough to filter chatter shorter than a few hundred microseconds — and settles cleanly into the tripped band only once the contact stays open.

> **What the model does not cover.** No trip module is chosen (memo 02's costed GAC SSW675 / Murphy HD9063 candidates survive costing but neither's aux-contact rating is pulled), so Rt=1.2 MΩ is this file's own INFERRED choice with no datasheet or industry EOL-loop convention behind it. The loop is high impedance (the same few-hundred-microamp class discrete_input.cir's divider runs at), not the low-Z, milliamp-class loop industrial supervised-contact practice normally uses — chosen to reuse the existing front-end rather than build a new one, and the cost is a genuinely tight margin: at the 6 V cranking dip with Rt at +5% tolerance, the tripped band's floor sits only 0.147 V above the wire-fault threshold, about half the nominal-case margin, before any harness leakage is stacked on top. This pin also needs to land on an ADC-capable channel, not a bare GPIO — a pin-mux consideration for whoever assigns the physical PTxx map.

PASS healthy (contact closed) band floor
PASS healthy band ceiling
PASS tripped (Rt in circuit) band floor, at 6V cranking dip
PASS tripped band ceiling, at 40V
PASS wire fault reads near zero
PASS tripped band never reaches the zener's 3.0V knee
PASS margin: wire-fault threshold to tripped floor (nominal Rt)
PASS margin: tripped ceiling to healthy floor (nominal Rt)

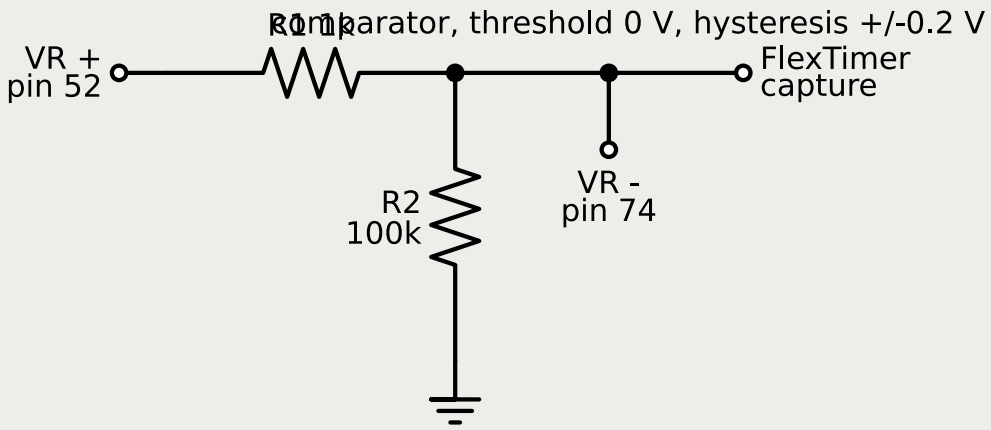
PASS ... both bands separated with real margin, not touching	wire-fault margi
PASS during bounce (t<4.5ms), reading never leaves the healthy band	
PASS settled value (t=100ms) approaches the DC asymptote (hand: 0.6206 V)	
PASS ... which sits inside the tripped band, not wire-fault or healthy	
PASS ... global minimum over the whole run stays clear of wire-fault	
PASS open-wire detection: needed here, unlike pins 20/24/71	no OEM harness to stay compatible with, and an undetectable open reads

Netlist: sim/blocks/trip_module_sense.cir

03 Speed and position

The most safety-critical input on the board.

Crank speed conditioner `pins 52, 74, 30` `7/7 checks`



A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

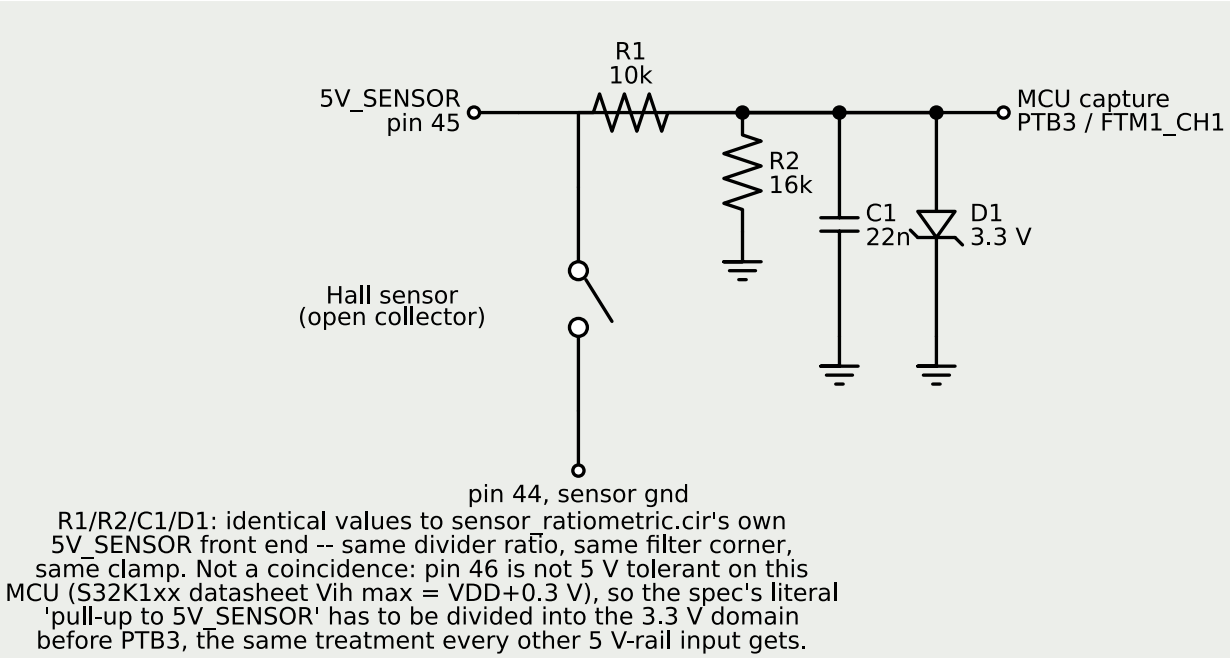
Why it looks like that. A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

> **What the model does not cover.** A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

PASS zero-cross, running (1500 rpm, 30 teeth)		30.000 edges
PASS zero-cross, cranking (150 rpm, 3 teeth)		3.000 edges
PASS fixed 5 V threshold, running		30.000 edges
PASS fixed 5 V threshold, cranking		0.000 edges
PASS ... so a fixed threshold never starts the engine	works at 1500 rpm, blind at 150 rpm -- passes on the bench	
PASS running tooth period		666.667 us
PASS ... jitter across the window		0.000 us

Netlist: sim/blocks/vr_conditioner.cir

Cam signal front-end `pins 45, 46, 44` `10/10 checks`



A cam Hall sensor's open-collector output, pulled up and divided into the 3.3 V domain before PTB3/FTM1_CH1, instead of the spec's literal pull-up straight to the 5 V sensor rail.

Why it looks like that. PTB3 is not 5 V tolerant for anything this design should rely on — the S32K148's own datasheet puts guaranteed valid-high recognition at VDD+0.3 V, about 3.6 V, and treats a higher DC injection allowance as a fault margin, not a design basis for a pin that sees 5 V every cam revolution for the machine's whole life. Nothing in sim/blocks/ ever modelled this pin before — there is a conditioner for the crank sensor and no cam equivalent — so the overvoltage sat uncaught. The divider (10k/16k) and filter (22 nF, 1176 Hz corner) reuse sensor_ratiometric.cir's own values exactly, since both share the same 5V_SENSOR rail and the same target domain: 94x above the cam's 12.5 Hz signal (half crank speed at 1500 rpm, one lobe per revolution assumed) and still well below switching-frequency noise.

> **What the model does not cover.** The real Hall sensor's saturation voltage, sink current, and internal pull-up (if any) are not modelled — no part chosen. Lobe count is INFERRED at 1 per cam revolution; the TOLERANCE section shows the filter margin survives a 3-lobe alternative just as well. A real cam signal's edge shape and any duty-cycle asymmetry from target

> geometry are not modelled, the same limitation vr_conditioner.cir already states for the crank signal's own non-ideal waveform.

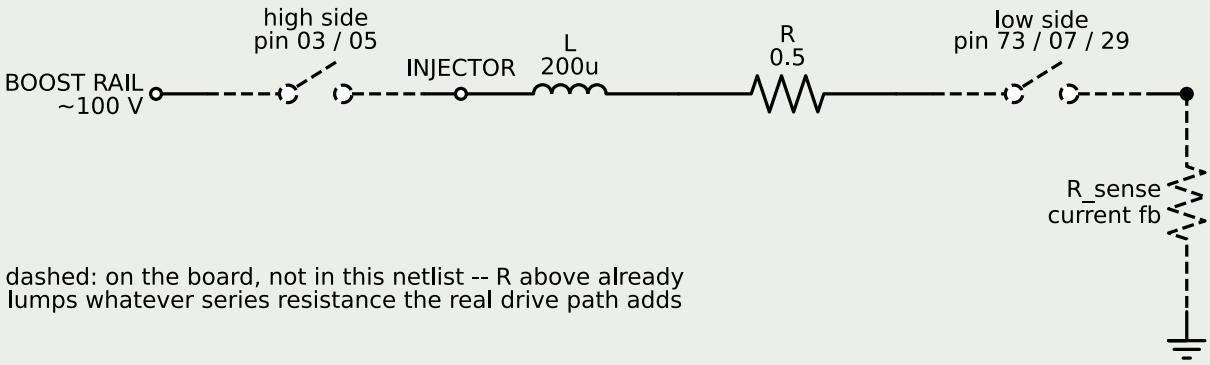
PASS sensor off, 5V_SENSOR nominal (5.0 V) → pad	3.072 V
PASS ... clears the 3.6 V abs max	0.528 V
PASS sensor_rail.cir's own fault floor (4.5 V) → pad	2.769 V
PASS ... still reads logic high (⩾2.31 V, 0.7*VDD)	2.769 V
PASS above-nominal rail (5.25 V) → pad	3.147 V
PASS ... still clears the 3.6 V abs max	0.453 V
PASS never exceeds 3.6 V abs max in transient	3.072 V
PASS rising edges, 170 ms window at 12.5 Hz (80 ms period)	2.000 edges
PASS 40 V harness short on signal line clamps pad to	3.308 V
PASS ... fault current limited to	3.669 mA

Netlist: sim/blocks/cam_frontend.cir

04 Injection

The most expensive stage on the board and the reason it exists.

Why the boost rail exists pins 03, 05 7/9 checks



The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

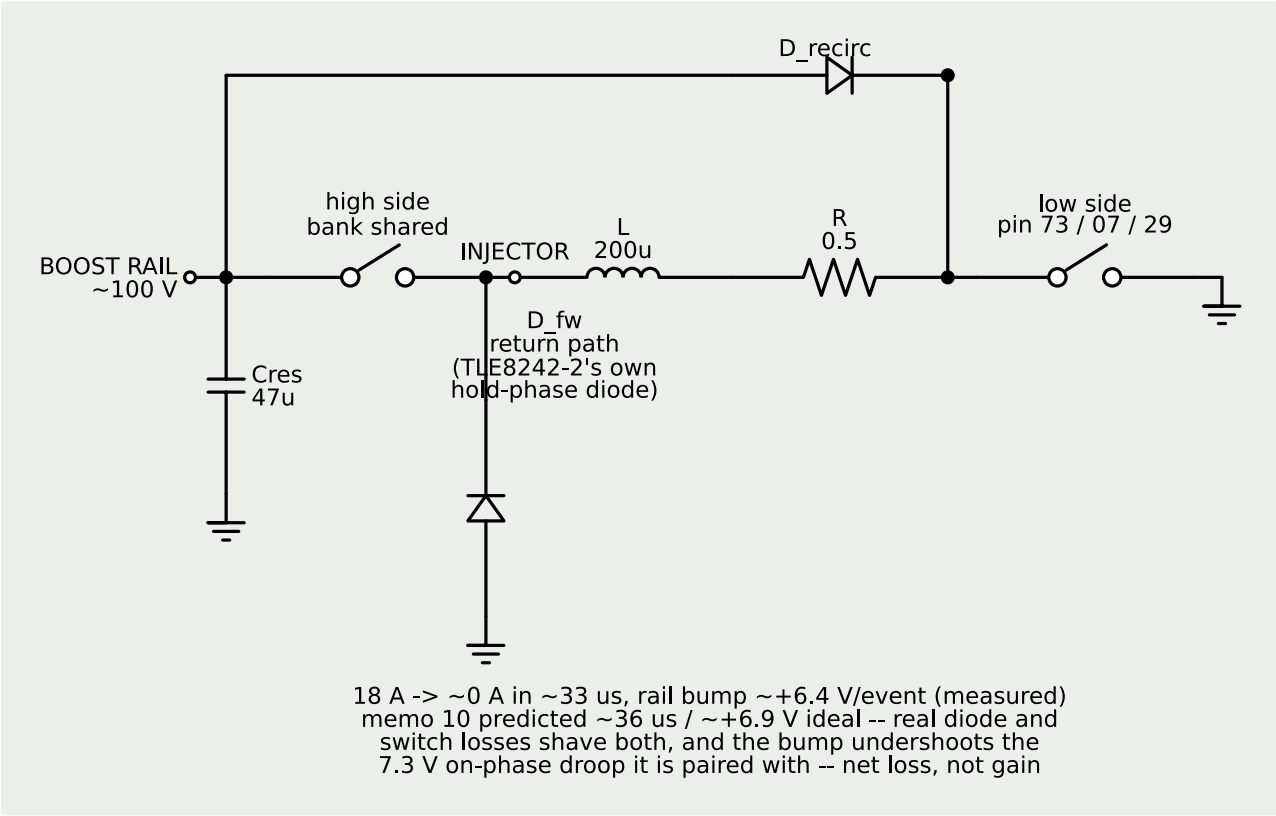
Why it looks like that. 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

> **What the model does not cover.** An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

PASS time to 18 A peak from 13.5 V battery	439.496 us
PASS time to 18 A peak from 100 V boost	37.796 us
PASS boost is faster by	11.628 x
PASS battery drive eats this much of a 1 ms injection	43.950 %
PASS boost drive eats this much of a 1 ms injection	3.780 %
FAIL ... boost is faster by, -40C ambient	9.856 x
PASS ... time to 18 A from 100 V boost, -40C ambient	37.296 us
FAIL ... boost is faster by, +125C ambient	19.017 x
PASS ... time to 18 A from 100 V boost, +125C ambient	38.496 us

Netlist: sim/blocks/injector_boost.cir

Turn-off / recirculation `pins 73, 07, 29` `15/15 checks`



The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47 uF) instead of a clamp that just burns it — plus a second diode (D_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

Why it looks like that. Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-"Questionable") and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33 us to collapse 18 A to zero (memo predicted ~36 us), 25x faster than freewheeling to the battery (830 us — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost_converter.cir's charge regulation (Bchga) has no bleed path either way; it just happens not to need one for the peak-only case.

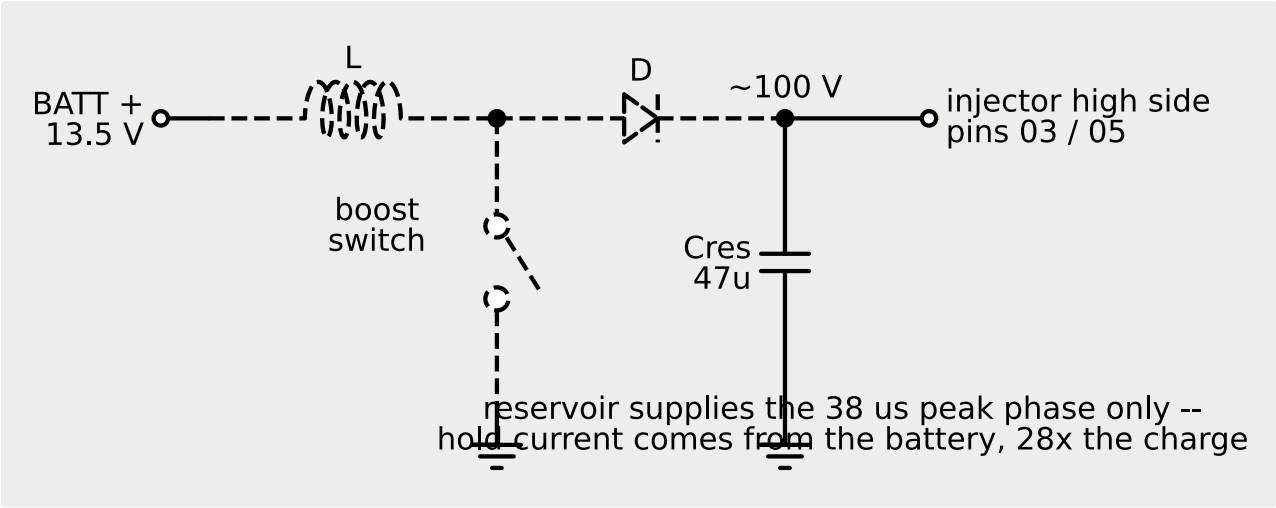
> **What the model does not cover.** The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any dV/dt read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery (Qrr), the diode's thermal transient, and — inherited from injector_boost.cir — magnetic saturation and needle motion.

PASS turn-off time, 18 A → ~0 A on the boost rail	
PASS ... vs freewheel-to-battery baseline (comparison only)	
PASS ... boost recirculation is faster by	
PASS on-phase droop, pilot event	
PASS ... matches boost_converter.cir's own droop figure	
PASS recovery bump, pilot event (isolated)	
PASS ... undershoots the droop it is paired with (real losses)	
PASS rail never exceeds 100 V setpoint (3-event circuit)	
PASS rail recovered by the next cylinder (26.67 ms after post)	
PASS hold-current cutoff into the SAME cap, no offsetting draw	
PASS ... THIS is the scenario that overshoots 100 V	
PASS ... so the verdict hinges on whether hold-current cutoffs share the boost-recirculation diode	peak-only recirculation never overshoots; hold-current reci
PASS peak instantaneous power in D_recirc during turn-off	
PASS ... exceeds the ~3.6 W worst-case AVERAGE by	

```
PASS differential-
mode edge vs                                  not modelled here (ideal switches) --
emi_filter.cir
```

Netlist: sim/blocks/injector_turnoff.cir

Boost rail reservoir `pins 03, 05` `7/7 checks`



The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

Why it looks like that. The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

> **What the model does not cover.** Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

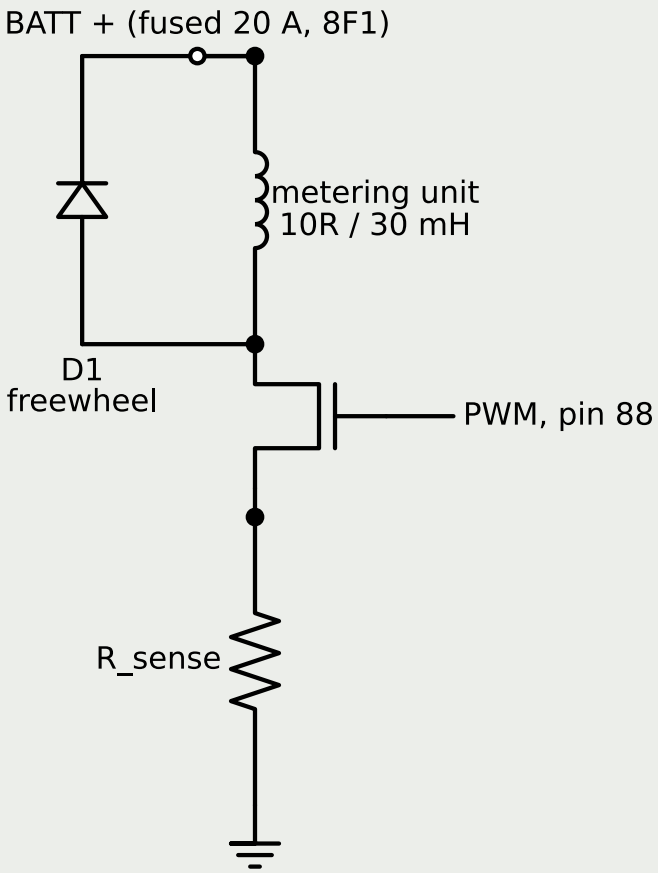
PASS droop, peak phase only (47 uF)		7.255 V
PASS ... rail stays usefully above battery		92.745 V
PASS droop, peak AND hold from the rail		210.913 V
PASS ... so the rail collapses	-111 V -- hold must come from the battery	
PASS hold phase costs this much more charge		29.071 x
PASS recovered by the next injection (26.67 ms)		100.000 V
PASS ... 50 mA average charging is enough		100.000 V

Netlist: sim/blocks/boost_converter.cir

05 Actuators

Outputs that move something.

Fuel metering unit driver `pin 88` `11/17 checks`



Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

Why it looks like that. Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

> **What the model does not cover.** The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

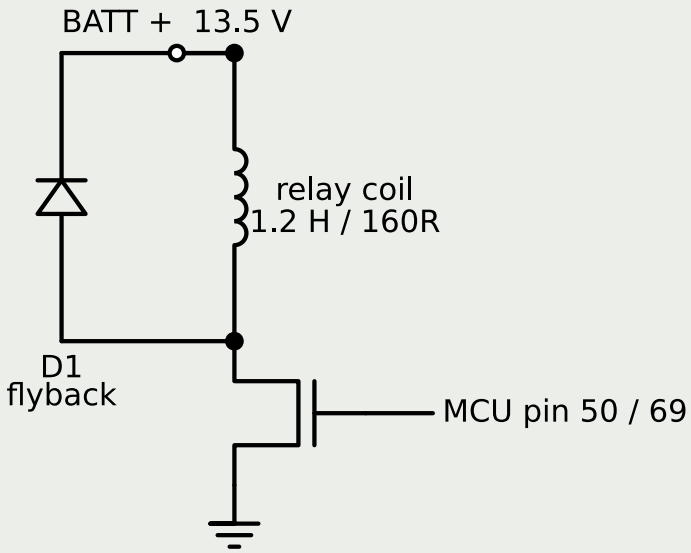
> rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

PASS mean current at 100 Hz		0.639 A
PASS mean current at 1 kHz		0.639 A
PASS mean current at 10 kHz		0.634 A
PASS ripple at 100 Hz		0.969 A
PASS ... as fraction of mean		151.786 %
PASS ripple at 1 kHz		0.118 A
PASS ripple at 10 kHz		0.012 A
PASS ripple scales as 1/f		9.974 x
PASS 100 Hz is not dither, it is chopping	152% ripple -- valve follows the PWM	
FAIL ... mean current at 100 Hz, -40C ambient		0.862 A
FAIL ... mean current at 1 kHz, -40C ambient		0.863 A
FAIL ... mean current at 10 kHz, -40C ambient		0.858 A
PASS ... control law: frequencies still agree, -40C		0.514 %
FAIL ... mean current at 100 Hz, +125C ambient		0.463 A
FAIL ... mean current at 1 kHz, +125C ambient		0.463 A
FAIL ... mean current at 10 kHz, +125C ambient		0.459 A
PASS ... control law: frequencies still agree, +125C		0.925 %

Netlist: sim/blocks/metering_unit_pwm.cir

Relay driver pins 50, 69

3/3 checks



A low-side FET switching a relay coil, run with and without the flyback diode.

Why it looks like that. With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

> **What the model does not cover.** The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

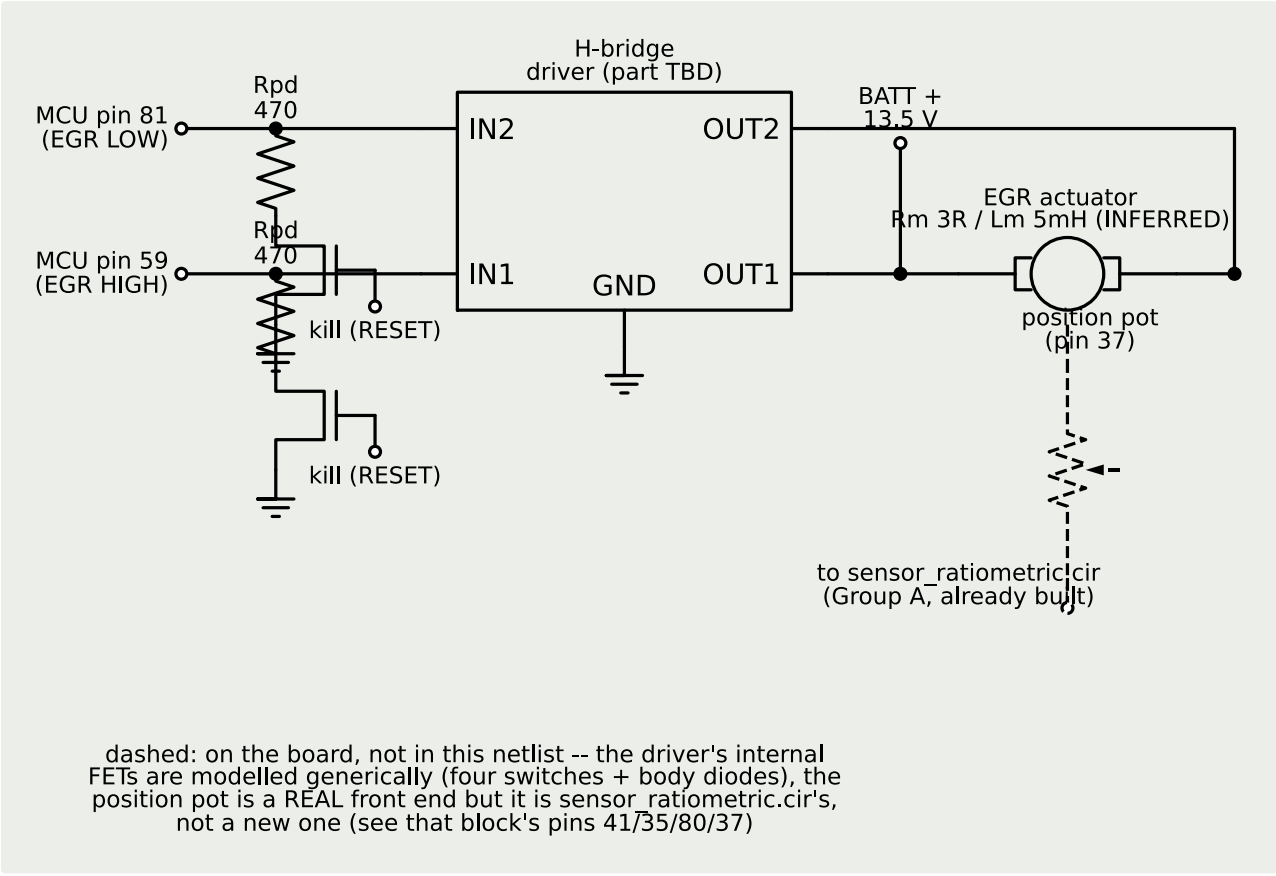
PASS drain peak WITH flyback diode	14.274 V
PASS drain peak WITHOUT diode	402,899.522 V
PASS ... diode suppresses the kick by	28,226.510 x

Netlist: sim/blocks/relay_driver.cir

EGR actuator H-bridge

pins 59, 81, position feedback pin 37

19/19 checks



Two logic/PWM lines into an external H-bridge driving a positional DC-motor actuator both ways — the one output on this board that is not a low-side switch — modelled as four generic switch positions with anti-parallel body diodes, plus the same active kill-clamp (Rpd 470 Ω, keyed to the supervisor's RESET) spec §4 requires at every driver gate.

Why it looks like that. This is a positional actuator, not an on/off solenoid: it is held at a commanded position, which is why it needs a bridge. Spec §4's outputs table lists pins 59/81 in the same row set as the six gates supervisor.cir already protects, but that file's own header enumerates only 88/73/07/29/03/05 — EGR was never added because this block did not exist. Reusing supervisor supervisor.cir's validated kill topology closes that gap rather than re-deriving it. The two-input "sign" scheme this design assumes (IN1 drives the S1+S4 diagonal, IN2 drives S3+S2) means no combination the ECU can present ever commands one leg's own high and low side together — real shoot-through is a property of the eventual driver IC's internal interlock, a part not yet chosen, not something these two lines can cause on their own. Stall current (jammed, or simply sitting at its own end of travel — normal operation for a positional actuator, not a fault) comes out to 4.36 A at an assumed 3 Ω armature resistance, comfortably past a class-typical 2–3 A small-

actuator driver rating: the eventual part needs active current limiting, not a fixed series resistor. Commanding coast with current already flowing is checked with and without the body diodes, the same comparison relay_driver.cir runs for its flyback diode: with them the bridge nodes stay within a diode drop of the rail; without them the ideal-switch model produces a multi-megavolt artifact, because the motor's stored energy has nowhere else to go.

> **What the model does not cover.** No H-bridge driver IC or FET is chosen — Ron, the body-diode figures, and the motor's Rm/Lm are all INFERRED class figures, the same gap as every other unselected FET on this board. Whether the eventual driver is integrated or discrete FETs behind a gate driver is undecided (memo 12 settled this for the injector and metering stages but not for EGR), so the 470 Ω kill-clamp value here is a uniform-standard choice, not a fresh Miller-coupling derivation for the 13.5 V domain EGR actually runs in. No mechanical model of the actuator exists — no return-spring constant, no gear ratio, no end-stop switches — so "coast lets the spring close the valve" is stated as the assumed fail-safe direction, not simulated. Position feedback on pin 37 is not modelled in this file at all: it is Group A ratiometric, already built and checked in sensor_ratiometric.cir, and building a second front end here would duplicate a proven circuit rather than reuse it.

PASS kill-clamp: gate below Vgsth (hand estimate 22.8 ns)		22.
PASS ... adversarial driver (100R/10V) still held below Vgsth		0
PASS ... adversarial settle voltage (hand: 10*4.95/104.95=0.471 V)		0
PASS stall current (Vemf=0, hand: 13.5/3.1)		4
PASS running current (Vemf=9V, hand: 4.5/3.1)		1
PASS coast current (IN1=IN2=0)		0
PASS stall current exceeds a class-typical 2-3 A driver rating		4
PASS ... so the eventual driver needs active current limiting, not a fixed series resistor	4.35 A at Rm=3 ohm alone exceeds a 2-3 A class	
PASS forward steady current at t=19.9ms matches section 2's stall figure		4
PASS WITH diodes: low side stays within a diode drop of ground (hand: -1.09 V)		-1
PASS WITH diodes: high side stays within a diode drop of the rail (hand: 14.59 V)		14
PASS WITHOUT diodes: low side swings far past ground		9,038,255
PASS WITHOUT diodes: high side swings far past the rail		9,038,269
PASS ... recirculation diodes are load-bearing, not decorative	bounded to 14.6 V with diodes vs 9.04e+06 V w	

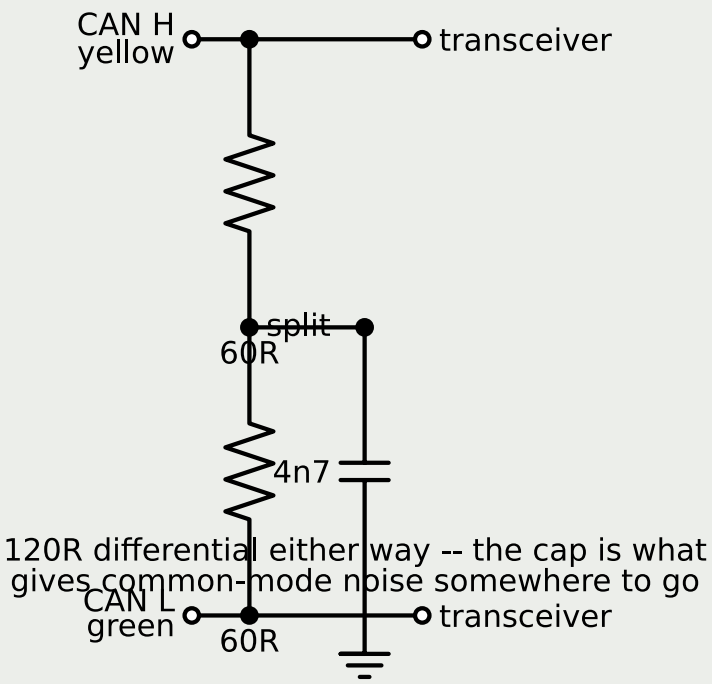
PASS position feedback (pin 37) is Group A ratiometric, proven in sensor_ratiometric.cir	0.308-2.769 V into the 3.3 V ADC, see check_sensor_ratiome
PASS rejected naive wiring: rail current with IN1=IN2 asserted	270
PASS ... matching the two-legs-of-2x50mOhm hand figure	270
PASS ADOPTED: decoded DIR/PWM driver, both inputs asserted	4
PASS ... so decode removes the fault state entirely	270 A naive vs 4.35 A decoded

Netlist: sim/blocks/egr_hbridge.cir

06 Communications

The bus to the genset controller.

CAN split termination `CAN H, CAN L` `7/8 checks`



A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

Why it looks like that. Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

> **What the model does not cover.** Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

PASS split termination, differential at 250 kbit/s	119.993 ohm
PASS single 120R, differential at 250 kbit/s	119.993 ohm
PASS ... identical differentially	0.000 ohm
PASS split termination, common-mode at 10 MHz	30.189 ohm
PASS single 120R, common-mode at 10 MHz	500.000 kohm
PASS ... split shunts common mode better by	16,562.490 x
PASS split common-mode at the bit rate stays high	138.099 ohm
FAIL ... at 200 ppm/C resistor tempco (matched direction, -40/125C)	123.952 ohm

Netlist: sim/blocks/can_termination.cir